



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Main)
UNIT TEST
30-05-2021

JEE(Main+Advanced) : NURTURE COURSE (PHASE : TEPS)

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on this Test Booklet.

Paper : Physics, Chemistry & Mathematics

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and –1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer and 0 in all other cases.
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited**.
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

Name of the Candidate (in Capitals) _____

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Topic :

PHYSICS

SEC-A : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:
Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : -1 If wrong option is selected.

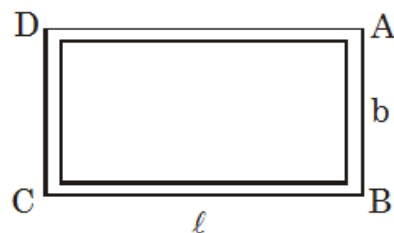
1. Experiment shows that two perfectly neutral parallel metal plates separated by a small distance d , attract each other via a very weak force, known as the Casimir force. The force per unit area of the plates, F , depends only on the Planck constant h , on the speed of light c , and on d . Which of the following has the best chance of being correct for F ? useful formula $E = hf$

(A) $F = \frac{hc}{d^2}$ (B) $F = \frac{hc}{d^4}$
 (C) $F = \frac{hd^2}{c}$ (D) $F = \frac{d^4}{hc}$

2. If $\vec{a} = 3\hat{i} + 4\hat{j}$ & $\vec{b} = 7\hat{i} + 24\hat{j}$ then the vector having the same magnitude as $\vec{b}$ and same direction as $\vec{a}$ is:

(A) $15\hat{i} + 20\hat{j}$
 (B) $10\hat{i} + 20\hat{j}$
 (C) $20\hat{i} + 15\hat{j}$
 (D) $15\hat{i} + 10\hat{j}$

3. In the figure shown ABCD is a rectangular smooth tube kept fixed in a vertical plane. A particle is projected from point A to reach point C with some speed. At the corners B and D velocity changes its direction by 90° without any change of its magnitude at that corner. If time taken on paths ABC and ADC are t_1 and t_2 respectively, then: (Given $\ell > b$)



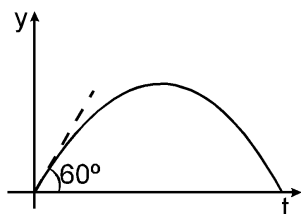
- (A) $t_1 = t_2$ (B) $t_1 > t_2$
 (C) $t_1 < t_2$ (D) none of these
4. Following is the cutting of a newspaper. Read it carefully and answer the question.

Car train mishap, one person dies.

"EMU local departing at 9:15 AM on its way from Hazrat Nizamuddin to Gaziabad meets with an accident near Sahibabad at a manless railway crossing. The accident occurred at 9:45 AM when the driver of the car tried to cross a railway crossing even when he saw the train coming. The driver of the car died in this accident. Without any delay the train continued for its destination, but due to damage in the engine, rest of the journey was travelled at $1/3^{\text{rd}}$ of its usual speed and it reached 2 hours late from its scheduled time even though it didn't stop anywhere in its rest of the journey". Assuming that it moves with constant speed everyday, find the time at which it reached Gaziabad that day.

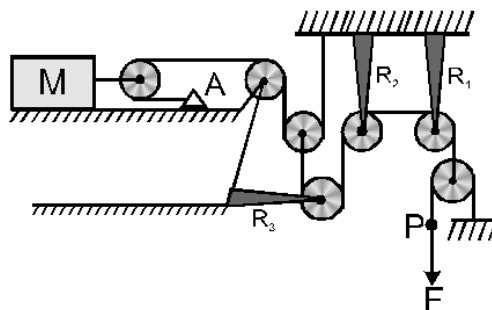
- (A) 12:15 pm (B) 12:30 pm
 (C) 12:45 pm (D) 1:15 pm

5. A particle is projected from the horizontal x - z plane, in vertical x - y plane where x -axis is horizontal and positive y -axis vertically upwards. The graph of ' y ' coordinate of the particle v/s time is as shown. The range of the particle is $\sqrt{3}$ m. Then the speed of the projected particle is:



- (A) $\sqrt{3}$ m/s
 (B) $\sqrt{\frac{403}{4}}$ m/s
 (C) $2\sqrt{5}$ m/s
 (D) $\sqrt{28}$ m/s
6. A particle is projected from a point P (2, 0, 0)m with a velocity 10 m/s making an angle 45° with the horizontal. The plane of projectile motion passes through a horizontal line PQ which makes an angle of 37° with positive x -axis, xy plane is horizontal. The coordinates of the point where the particle will strike the line PQ is: (Take $g = 10 \text{ m/s}^2$)
- (A) (10, 6, 0)m
 (B) (8, 6, 0)m
 (C) (10, 8, 0)m
 (D) (6, 10, 0)m

7. In the given arrangement, mass of the block is M and the surface on which the block is placed is smooth. Assuming all pulleys to be massless and frictionless, strings to be inelastic and light, R_1 , R_2 and R_3 to be light supporting rods, then acceleration of point 'P' will be (A is fixed) :

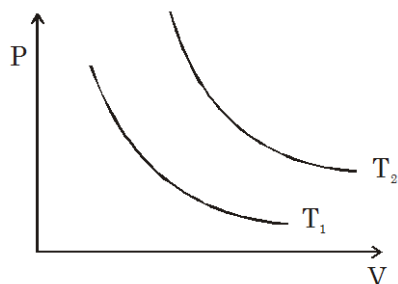


- (A) 0
 (B) ∞
 (C) $\frac{4F}{m}$
 (D) $\frac{2F}{m}$
8. If I_1 is the moment of inertia of a thin rod about an axis perpendicular to its length and passing through its centre of mass and I_2 is the moment of inertia of the ring about an axis passing through its centre and perpendicular to its plane formed by bending this rod to the ring shape. Then
- (A) $I_1 : I_2 = 1 : 1$
 (B) $I_1 : I_2 = \pi^2 : 3$
 (C) $I_1 : I_2 = \pi : 4$
 (D) $I_1 : I_2 = 3 : 5$
9. A hoop of radius r weighs m kg. It rolls along a horizontal floor so that its centre of mass has a speed v m/s. How much work has to be done to stop it?
- (A) $\frac{1}{2}mv^2$
 (B) mv^2
 (C) $2mv^2$
 (D) $\frac{1}{3}mv^2$

10. The volume of a mixture of ice and water is found to decrease by $0.125 \times 10^{-6} \text{ m}^3$ without change in temperature when $10 \times 10^{-3} \text{ kg}$ of a metal at 100°C is immersed inside it. The relative density of ice is 0.917. What is the specific heat of metal, if latent heat of ice = $336 \times 10^3 \text{ J/kg}$?

(A) $264 \text{ J kg}^{-1} \text{ K}^{-1}$
 (B) $364 \text{ J kg}^{-1} \text{ K}^{-1}$
 (C) $464 \text{ J kg}^{-1} \text{ K}^{-1}$
 (D) $564 \text{ J kg}^{-1} \text{ K}^{-1}$

11. Two isotherms are shown on p-V diagram. If the mass of gas is same in both cases, which one of the following inferences is correct?



(A) $T_1 = T_2$ (B) $T_1 \cdot T_2 = 1$
 (C) $T_1 > T_2$ (D) $T_1 < T_2$

12. A calorimeter contains 50 gms of water at 50°C . The temperature falls to 45°C in 10 minutes. When the calorimeter contains 100 gms of water at 50°C it takes 18 minutes for the temperature to become 45°C . What is the water equivalent of the calorimeter?

(A) 10 gm (B) 12.5 gm
 (C) 5 gm (D) 15 gm

13. A mass of 2 kg is suspended from a fixed point by a wire of length 3 metres and diameter 0.5 mm. Initially the wire is just unstressed, the mass resting on a fixed support. By how much the temperature (approx.) should fall if the mass is to be entirely supported by wire? (Young's modulus of wire 206 GPa and $\alpha = 11 \times 10^{-6}/^\circ\text{C}$).

(A) 20°C
 (B) 44°C
 (C) 30°C
 (D) 24°C

14. Calculate the internal energy of air (diatomic gas) in a room of volume 100 m^3 at atmospheric pressure (10^5 N/m^2).

(A) $2.5 \times 10^7 \text{ J}$
 (B) $1.5 \times 10^7 \text{ J}$
 (C) $2 \times 10^7 \text{ J}$
 (D) 10^7 J

15. A gas held in a container enclosed by a piston undergoes expansion such that the rate of collision of molecules among themselves remains constant. The process is most appropriately represented by:

(A) $PV^2 = \text{constant}$
 (B) $PV = \text{constant}$
 (C) $PV^{-1} = \text{constant}$
 (D) $PV^{1/2} = \text{constant}$

16. A planet radiates heat at a rate proportional to the fourth power of its surface temperature T . If such a steady temperature of the planet is due to an exactly equal amount of heat received from the Sun, then which of the following statement is true?

- (A) The planet's surface temperature varies inversely as the distance of the Sun.
 (B) The planet's surface temperature varies directly as the square of its distance from Sun.
 (C) The planet's surface temperature varies inversely as the square root of its distance from Sun.
 (D) The planet's surface temperature is proportional to fourth power of its distance from the Sun.

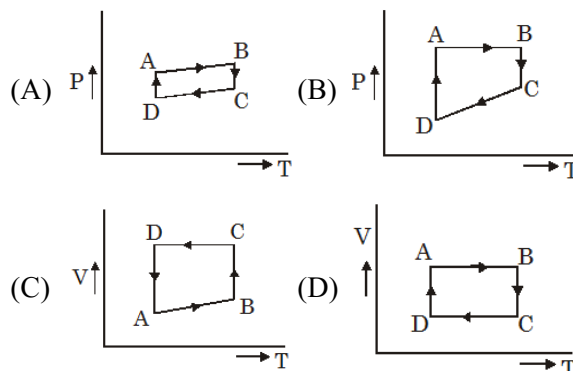
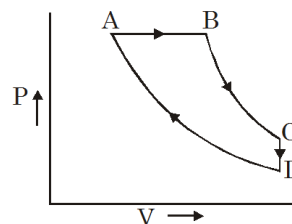
17. During an adiabatic process the pressure of a gas is found to be proportional to the cube of the absolute temperature. The ratio of the specific heats of the gas is

- (A) 2 (B) $\frac{3}{2}$
 (C) $\frac{5}{3}$ (D) $\frac{4}{3}$

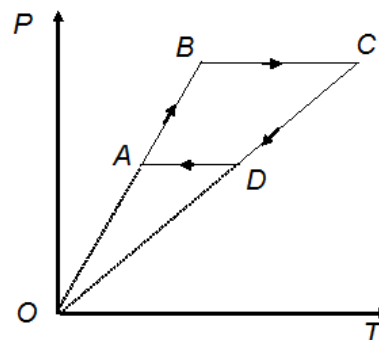
18. A rod of length ℓ with thermally insulated lateral surface is made of a material whose thermal conductivity K varies as $K = C/T$, where C is a constant. The ends are at temperatures T_1 and T_2 . Find the heat flow density (Rate of heat flow per unit area of the rod).

- (A) $C \log \frac{T_2}{T_1}$ (B) $\frac{C}{\ell} \log \left(\frac{T_2}{T_1} \right)$
 (C) $\frac{C}{\ell} \log (T_1 T_2)$ (D) $C \ell \log \left(\frac{T_2}{T_1} \right)$

19. A cyclic process ABCD is shown in the P-V diagram. Which of the following curves represent the same process?



20. Three moles of an ideal monoatomic gas undergoes the process ABCDA as shown, where $T_A = 300\text{K}$, $T_B = 600\text{K}$, $T_C = 1800\text{K}$ and $T_D = 900\text{K}$. Find the work done by the gas during the cycle.



- (A) 17.8 kJ (B) 14.9 kJ
 (C) 19.9 kJ (D) 21.1 kJ

SEC-B : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

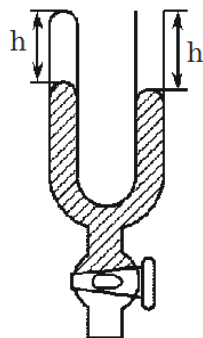
Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

1. Two fixed points A and B are 20 metres apart. At time $t = 0$, the distance between a third point C and A is 20 metres and the distance between C and B is 10 metres. The component of velocity of point C along both CA and CB at any instant is 5m/s. Find the distance (in m) between A and C at the instant all the three points are collinear.
2. A particle with a mass of $M = 10$ kg is suspended in an elevator with the help of three identical light extensible strings along the same vertical line. One of them is tied to the elevator ceiling the other two are tied to the floor. The ropes are stretched so that when the elevator is at rest, the force of tension in each of the lower ropes is 5 N. Find the tension (in N) in the upper rope when the acceleration of the elevator is 3 m/s^2 directed up. Take $g = 10 \text{ m/s}^2$.

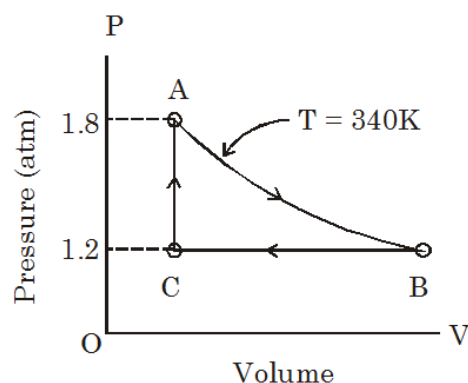
3. A man weighing 50 kg standing at the edge of a platform of mass 100 kg rotating at 20RPM moves towards its centre. If the radius of the platform is 1.5 m, what is the work done (in J) by the man? take $\pi^2 = 10$ and round off to nearest integer.
4. The ends of a copper rod of length 1 m and area of cross-section 1 cm^2 are maintained at 0°C and 100°C . At the centre of the rod there is a source of heat of power 25 W. Calculate the temperature gradient (in $^\circ\text{C m}^{-1}$) in the colder half of rod in the steady state. Thermal conductivity of copper is $400 \text{ W m}^{-1} \text{ K}^{-1}$.
5. The pressure of a gas in a 100 ml container is 200 kPa and the average translational kinetic energy of each gas particle is $6 \times 10^{-21} \text{ J}$. Find the number of gas particles in the container. (in multiples of 10^{21})
6. Find the volume (in mm^3) of a mercury thermometer bulb, if you know that at 0°C mercury fills only the bulb completely and between divisions 0 and 100°C , the volume of channel at 0°C is $V = 3 \text{ mm}^3$. Coefficient of volume expansion of mercury $= 1.8 \times 10^{-4} \text{ K}^{-1}$, the linear coefficient expansion of the glass is $8 \times 10^{-6} \text{ K}^{-1}$. Round off to nearest integer.
7. Water may be supercooled to a temperature $t = -10^\circ\text{C}$. Such state is unstable, and any perturbation of water transforms it into ice with temperature to $= 0^\circ\text{C}$. What is the mass (in gm) of ice formed from supercooled water, if its mass is 1 kg? The specific heat capacity of water is $4.2 \text{ kJ / (kg} \cdot \text{K)}$. Specific heat of melting ice is 0.33 MJ / kg . Round off to nearest integer.

8. Mercury is poured in a uniform U tube sealed on one side and open on another. It is seen that the level of mercury in the open side and closed side is both $h = 20$ cm. At the bottom end, the U tube is connected to a valve as shown. When the valve is opened, the level of mercury in the sealed side drops by 10 cm. By what distance (in cm) has the level fallen in the open end? The density of mercury is $13.6 \times 10^3 \text{ kg/m}^3$. Atmospheric pressure $p_0 = 0.1 \text{ MPa}$. Round off to nearest integer



9. Calculate the change in entropy (in J/K) of the universe if 0.1 kg of water at 87°C is mixed with 0.2 kg of water at 27°C in a thermally insulated container. (Assume $\ln(1+x) = x$ if $x \ll 1$)

10. A 1 mol sample of an ideal gas for which $c_v = 3R/2$ follows the closed thermodynamic cycle shown in figure. There are three legs in the cycle: an isothermal expansion, A to B; an isobaric compression, B to C; and a constant-volume increase in pressure, C to A.



Find the efficiency of the cycle in percentage.

Take $\ln 1.5 = 0.40$. Round off to nearest integer.

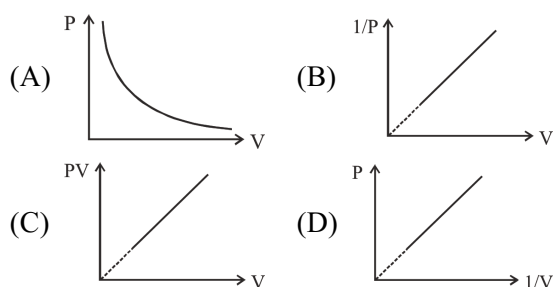
Topic :

CHEMISTRY

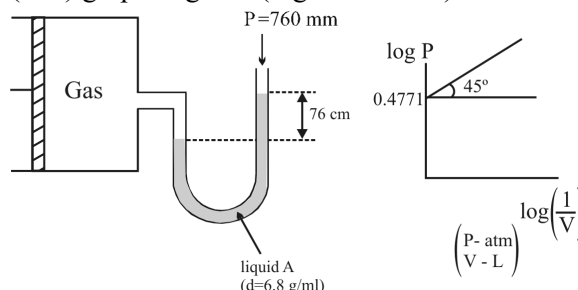
SEC-A : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:
Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : -1 If wrong option is selected.

- A 5000 W gas discharge lamp emits 5.0 W of UV radiation in narrow range centered around 198 nm wavelength. Then number of photons of this wavelength emitted in one minute :
 (Given : $h = 6.6 \times 10^{-34}$ J-S, $C = 3 \times 10^8$ m/s)
 (A) 2×10^{16} (B) 3×10^{20}
 (C) 5×10^{18} (D) 8.3×10^{16}
- The temperature at which 2nd Virial coefficient of a Vander Waal's gas is zero is -
 (A) Critical temperature
 (B) Boiling temperature
 (C) Boyle's temperature
 (D) None of these
- Which of the following curves does not represent Boyle's Law -



- For an ideal gas kept in a container attached with manometer as shown, find volume of container & for the given gas $\log P$ vs $\log (1/V)$ graph is given ($\log 3 = 0.4771$)



- (A) 3 L (B) 1.5 L
 (C) 4.5 L (D) 2 L
- A certain transition in H-spectrum from an excited state to ground state in one or more steps gives rise to total 10 different lines. How many of them will belong to UV spectrum -
 (A) 3 (B) 4 (C) 5 (D) 6
 - Calculate the mass of anhydrous oxalic acid, which can be oxidised to CO_2 (g) by 100 ml of an MnO_4^- solution, 10 ml of which is capable of oxidising 50 ml of 1N I^- to I_2
 (A) 45 gm (B) 22.5 gm
 (C) 30 gm (D) 12.25 gm
 - V_1 mL of NaOH solution of normality X is mixed with V_2 mL of $\text{Ba}(\text{OH})_2$ solution of normality Y. The mixture is completely neutralised by 100 mL of 0.1 N HCl. If $\frac{V_1}{V_2} = 1/4$ and $\frac{x}{y} = 4$, what fraction of the acid is neutralise by $\text{Ba}(\text{OH})_2$:-
 (A) 0.5 (B) 0.25 (C) 0.33 (D) 0.67

8. Consider the following compounds.

- (i) IF_5 (ii) ClF_4^- (iii) XeO_2F_2 (iv) NH_2^-
 (v) BCl_3 (vi) BeCl_2 (vii) AsCl_5 (viii) BO_3^{3-}
 (ix) NO_2^-

	List-I (Species in which central atom uses valency shell orbital)		List-II (Number of species out of given above)
P	All three p-orbitals are involved in hybridisation of central atom	1	One
Q	Only two p-orbitals are involved in hybridisation of central atom	2	Five
R	Only one p-orbital is involved in hybridisation of central atom	3	Three
S	Atleast one d-orbital is involved in hybridisation of central atom	4	Four

- (A) P-2 Q-3 R-1 S-4
 (B) P-2 Q-4 R-1 S-3
 (C) P-4 Q-3 R-2 S-1
 (D) P-4 Q-2 R-1 S-3

9. Consider the following table regarding interhalogen (when two different halogens react with each other, interhalogen compounds are formed)

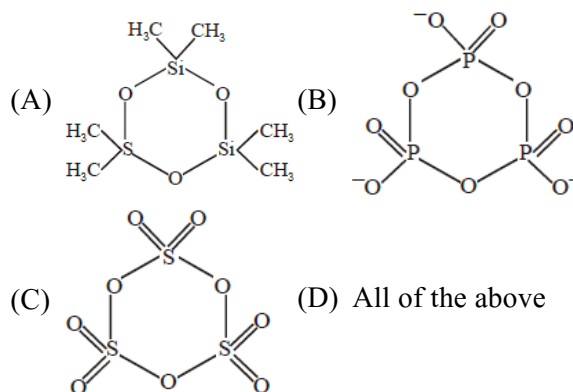
XY_n (where Y is more electronegative than X)

Polarity	Planarity	Total number of d-orbitals used in hybridisation of central atom	Value of n for respective interhalogen
Non-polar	Non-planar	R_1	A
Polar	Non-planar	R_2	B
Polar	Planar	R_3	C

Select the CORRECT statement about table :

- (A) Value of R_1 is 3 and value of A is 7
 (B) Value of R_2 is 1 and value of B is 3
 (C) Value of R_3 is 2 and value of C is 5
 (D) All are correct

10. How many compound are isoelectronic with $\text{Si}_3\text{O}_9^{6-}$ type of cyclic silicate



List- I		List- II	
(P)	$\text{PCl}_5(\text{g})$	(1)	involves d-orbital in hybridisation
(Q)	C_6H_6	(2)	species has atleast 4 atoms in one plane
(R)	SO_3	(3)	species has atleast one $\text{p}\pi\text{-d}\pi$ type of π -bond
(S)	SO_2Cl_2	(4)	non-planar species
		(5)	species has all equal bond length

The correct option is

- (A) $\text{P} \rightarrow 2,3,5$; $\text{Q} \rightarrow 1$; $\text{R} \rightarrow 3,4$; $\text{S} \rightarrow 1,4,5$
 (B) $\text{P} \rightarrow 1,2,3$; $\text{Q} \rightarrow 1$; $\text{R} \rightarrow 3,4$; $\text{S} \rightarrow 1,4,5$
 (C) $\text{P} \rightarrow 1,2,4$; $\text{Q} \rightarrow 2$; $\text{R} \rightarrow 2,3$; $\text{S} \rightarrow 3,4$
 (D) $\text{P} \rightarrow 1,2,3$; $\text{Q} \rightarrow 1,3$; $\text{R} \rightarrow 1,2,4$; $\text{S} \rightarrow 1,4,5$

12. Which of the following order is correct.

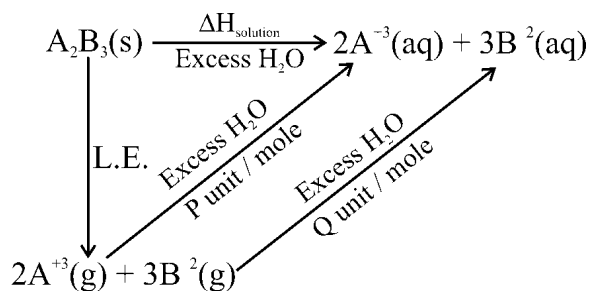
- (A) $\text{CO}_2 < \text{CO}_3^{2-}$ ($\text{O}-\hat{\text{C}}-\text{O}$ bond angle)
 (B) $\text{NH}_2^\ominus > \text{NH}_3$ ($\text{H}-\hat{\text{N}}-\text{H}$ bond angle)
 (C) $\text{BeCl}_2 < \text{SnCl}_2$ (dipole moment)
 (D) $\text{H}_2\text{O} < \text{H}_2\text{O}_2$ (Strength of Hydrogen bonding)

13. In which of the following species presence of double bond does not affect idealised bond angle :-

- (I) POCl_3 (II) XeO_3F_2 (III) XeO_2F_4 (IV) XeO_4
 (A) II, III, IV (B) I, II only
 (C) III, IV only (D) I, II, III

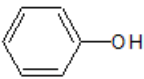
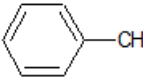
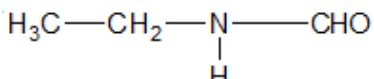
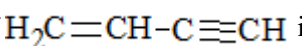
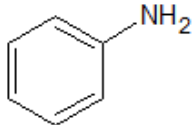
14. In the given cycle, if P and Q are the magnitude of hydration energies (H.E.) per mole for $\text{A}^{+3}(\text{g})$ and $\text{B}^{-2}(\text{g})$ ions respectively then the value of $\left(\text{P} + \frac{3}{2}\text{Q}\right)$ will be.

(Consider only magnitude of H.E.), (L.E. = Lattice energy of Ionic compound)

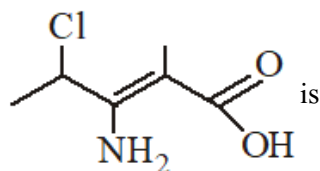


- (A) $\frac{\text{L.E.} - \Delta\text{H}_{\text{sol}}}{2}$
 (B) $\frac{3}{2}(\Delta\text{H}_{\text{sol.}} - \text{I.E.})$
 (C) $\Delta\text{H}_{\text{sol.}} - \text{L.E.}$
 (D) $\Delta\text{H}_{\text{sol.}} + \text{L.E.}$

15. Choose the correct option

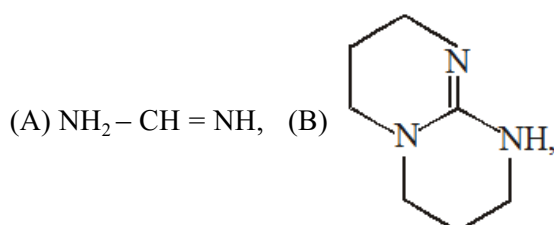
- (A)  and  are homologue
 IUPAC name of  is N-ethyl ethanamine
 (B) IUPAC name of  is but-1-en-3-yne
 (C)  is heterocyclic compound
 (D)

16. The IUPAC name of the compound



- (A) 3-amino-4-chloro-2-methylpent-2-enoic acid
 (B) 2-methyl-3-amino-4-chloro-2-pentenoic acid
 (C) 1-hydroxy-1-oxo-2-methyl-3-amino-4-chloro-2-pentene
 (D) 3-amino-2,4-dimethyl-4-chloro-2-butenic acid

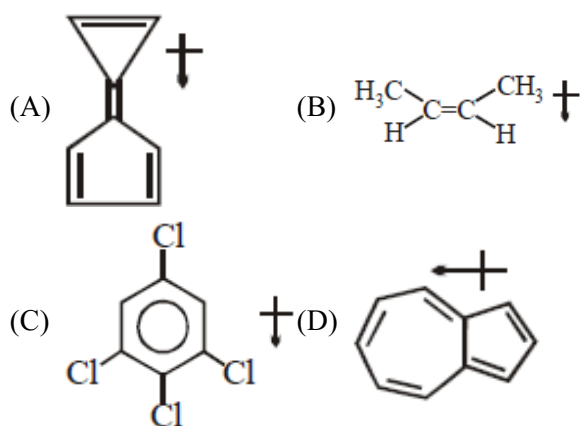
17. The **increasing order** of pK_b for the following compounds will be :



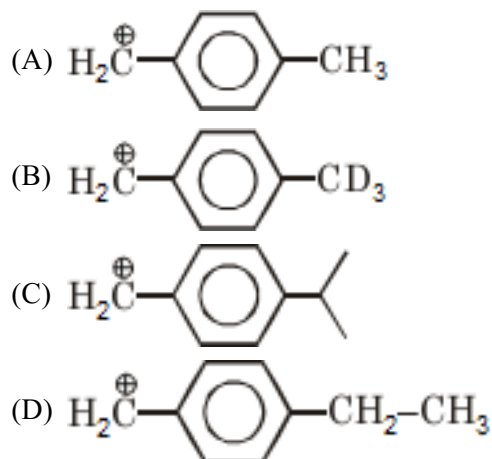
(C) CH_3NHCH_3

- (A) (A) < (B) < (C) (B) (C) < (A) < (B)
 (C) (B) < (A) < (C) (D) (B) < (C) < (A)

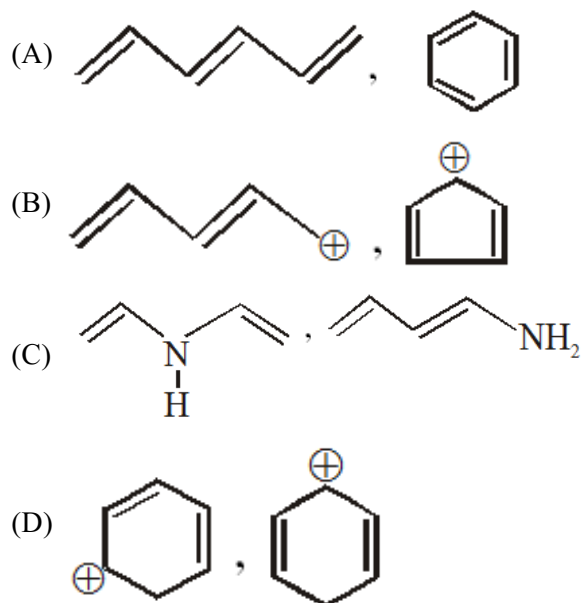
18. Which of the following compound has incorrect direction of net dipole moment shown ?



19. Which of the following carbocation is of highest stability ?



20. In which of the following pairs first one is having more resonance energy than the second one -



SEC-B : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

1. A valve between a 5 litre tank in which the gas pressure is 9 atm and a 10 litre tank containing gas at 6 atm is opened and pressure equilibration ensures at a constant temperature. What is the final pressure in the two tanks.
2. A mixture of NaI and CaBr₂ having mole fraction of CaBr₂ equal to $\frac{1}{3}$. To this mixture water is added and solution is made upto 1 litre where molality of NaI is found to be $\frac{10}{9}$. If density of solution is 1.15 g/ml. Calculate mass (in gm) of NaI in original mixture. (Na = 23, Ca = 40, Br = 80, I = 127)
3. 1 litre solution containing 0.1 M Na₂S₂O₃ & 0.2 M IO₄⁻ is titrated with I₂ according to following reaction.

$$\text{Na}_2\text{S}_2\text{O}_3 + \text{I}_2 \rightarrow \text{Na}_2\text{S}_4\text{O}_6 + \text{I}^-$$

$$\text{IO}_4^- + \text{I}_2 \rightarrow \text{IO}_3^-$$
 Calculate millimoles of I₂ consumed?
Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

4. If value of spin quantum number(s) = -1/2, 0, +1/2 then calculate number of groups in the new form of periodic table if all other rules of electronic configurations are remain same.

5. Consider the following table

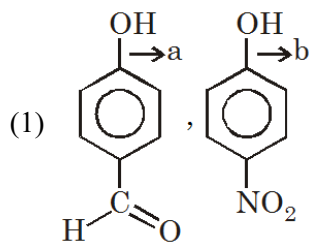
Compounds (X are monovalent surrounding atoms)	Central atoms (A to D) belong to group	Characteristics of compounds	Number of lone pair(s) at central atom
(i) AX _{n1}	16	Planar and polar	m ₁
(ii) BX _{n2}	15	Trigonal pyramidal	m ₂
(iii) CX _{n3}	14	Zero dipole moment	m ₃
(iv) DX _{n4}	13	All X-D-X bond angle are 120°	m ₄

Then calculate value of expression

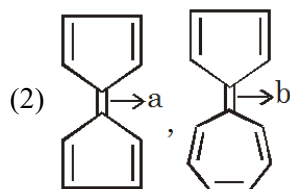
$$\left| \frac{n_1 + n_2 + n_3 + n_4}{m_1 + m_2 + m_3 + m_4} \right|^2$$

6. Count the number of incorrect statement among the following :-
 (A) The first ionisation potential of Al is less than the first ionisation potential of Mg
 (B) Radius of hydrated Li⁺ is more than that of hydrated Cs⁺
 (C) The formation of S²⁻ is an endothermic process
 (D) The first ionisation energies (in kJ mol⁻¹) of carbon, silicon, germanium, tin, and lead are 1086, 786, 761, 708 and 715 respectively :
 (E) Down the group, electronegativity decreases from B to Tl
 (F) Among oxides, CO is neutral, GeO is acidic and SnO is amphoteric.
 (G) The 4f-and 5f-inner transition elements are placed separately at the bottom of the periodic table to main its structure.

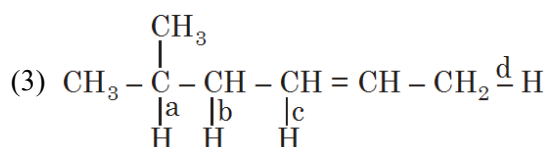
7. How many properties given are **CORRECT** :



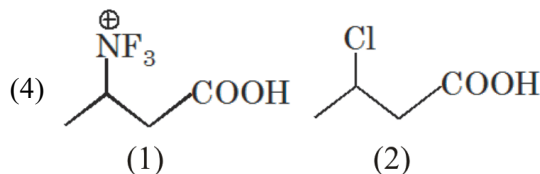
Bond length : $a > b$



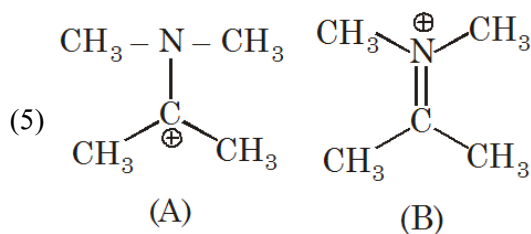
Rotational energy barrier : $a > b$



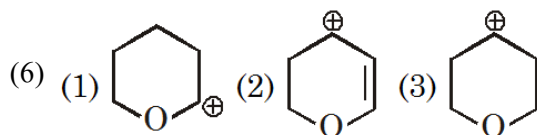
C-H bond energy : $c > b > a > d$



Acidic strength : $2 > 1$

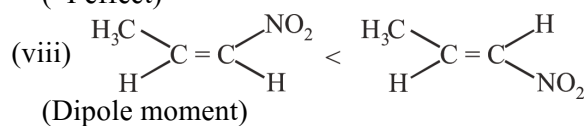
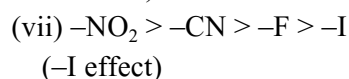
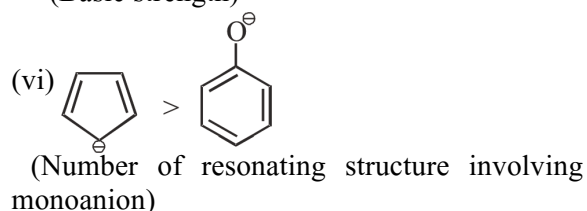
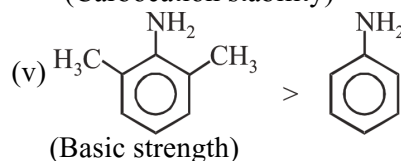
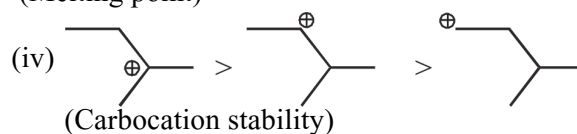
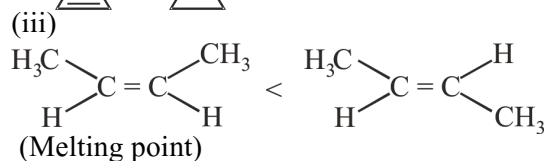
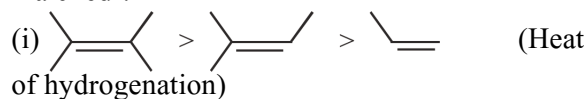


Stability : $B > A$

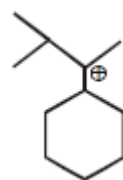


Stability $1 > 2 > 3$

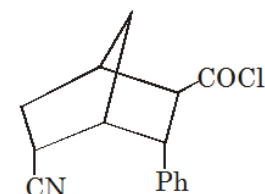
8. How many of the following orders are correctly matched ?



9. How many hyperconjugation structures are possible for following compound which involve C-H σ -bond :-



10. Degree of unsaturation of following compound is :



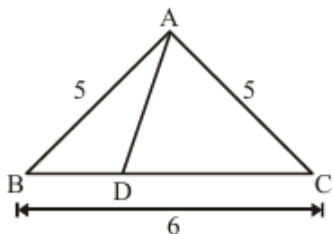
Topic :

MATHEMATICS

SEC-A : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:
Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : -1 If wrong option is selected.

1. Figure shows triangle ABC such that $AB = AC = 5$, $BC = 6$. Point D is on BC such that $\frac{BD}{DC} = \frac{1}{2}$, then length of side AD is :-



- (A) 4 (B) $\sqrt{13}$ (C) $\sqrt{15}$ (D) $\sqrt{17}$
2. If $\sin^2 x - (c-1)\sin x + 2c - 6 \geq 0$ for every $x \in \mathbb{R}$ then the set of values of c is
- (A) $[2, \infty)$ (B) $[4, \infty)$
 (C) $(-\infty, -2)$ (D) $(-\infty, -4]$
3. If $y^2 + 2y + \sec^2\left(\frac{\pi}{2}(y+x)\right) = 0$, then which of the following holds good?
- (A) $y = 1, x \in \mathbb{I}$
 (B) $y = -1, x$ is odd integer
 (C) $y \in \mathbb{R}, x \in \emptyset$
 (D) $y = -1, x$ is even integer

4. If $x = a(b-c)$, $y = b(c-a)$, $z = c(a-b)$ then the value of $\left(\frac{x}{a}\right)^3 + \left(\frac{y}{b}\right)^3 + \left(\frac{z}{c}\right)^3$ is:
- (A) $\frac{2xyz}{abc}$ (B) $\frac{xyz}{abc}$
 (C) 0 (D) $\frac{3xyz}{abc}$
5. A is a point on either of two rays $y + \sqrt{3}|x| = 2$ at a distance of $\frac{4}{\sqrt{3}}$ units from their point of intersection. The co-ordinates of the foot of perpendicular from A on the bisector of the angle between them are
- (A) $\left(-\frac{2}{\sqrt{3}}, 2\right)$ (B) (0, 0)
 (C) $\left(\frac{2}{\sqrt{3}}, 2\right)$ (D) (0, 4)
6. Find the locus of the mid point of the chord of a circle $x^2 + y^2 = 4$ such that the segment intercepted by the chord on the curve $x^2 - 2x - 2y = 0$ subtends a right angle at the origin.
- (A) $x^2 + y^2 - 2x - 2y = 0$
 (B) $x^2 + y^2 + 2x - 2y = 0$
 (C) $x^2 + y^2 + 2x + 2y = 0$
 (D) $x^2 + y^2 - 2x + 2y = 0$
7. If sides of $\triangle ABC$ are $x - 2y = 7$, $2x + y = 4$ and $y = 4$, then slope of line joining orthocentre and circumcentre of $\triangle ABC$ is-
- (A) $\frac{4}{3}$ (B) $\frac{9}{8}$ (C) 1 (D) $\frac{7}{8}$

8. The line $2x - y = 3$ is rotated through an angle $\frac{\pi}{4}$ in anticlockwise direction about the point $(2, 1)$, then equation of line in its new position is
- (A) $3x + y = 7$
 (B) $3x - y = 7$
 (C) $3x + y = 5$
 (D) $3x - y = 5$
9. Centre of the circle which intersects each of the following three circles
 $x^2 + y^2 + 2x + 17y + 4 = 0$
 $x^2 + y^2 + 7x + 6y + 11 = 0$
 $x^2 + y^2 - x + 22y + 3 = 0$
 orthogonally is
- (A) $(3, 2)$ (B) $(-1, 2)$
 (C) $(2, 3)$ (D) $(2, -1)$
10. If $3\ell^2 + 6\ell + 1 = 6m^2$ then equation of circle for which $\ell x + my + 1 = 0$ is always a tangent, is
- (A) $(x + 3)^2 + y^2 = 6$
 (B) $(x - 3)^2 + y^2 = 6$
 (C) $x^2 + (y - 3)^2 = 6$
 (D) $x^2 + (y + 3)^2 = 6$
11. Let a point P is lying on x-axis, such that tangents PT_1 & PT_2 are drawn to the circle $x^2 + y^2 = 16$. If rhombus PT_1QT_2 is completed, then area of the rhombus is (Where Q lies on the circle)
- (A) $24\sqrt{3}$ (B) $22\sqrt{3}$
 (C) $22\sqrt{2}$ (D) $24\sqrt{5}$
12. Two circles are given as $x^2 + y^2 + 14x - 6y + 40 = 0$ and $x^2 + y^2 - 2x + 6y + 7 = 0$ with their centres as C_1 and C_2 respectively. If equation of another circle whose centre C_3 lies on the line $3x + 4y - 16 = 0$ and touches the circle (whose center is C_1) externally and also $C_1C_2 + C_2C_3 + C_3C_1$ is minimum is $x^2 + y^2 + ax + by + c = 0$ then the value of $(a + b + c)$ is
- (A) 2 (B) 8 (C) 16 (D) 0
13. Two circles C_1 and C_2 both passes through the points $A(1, 2)$ and $E(2, 1)$ and touch the line $4x - 2y = 9$ at B and D respectively. The possible coordinates of a point C such that the quadrilateral ABCD is a parallelogram is
- (A) $(2, -1)$ (B) $(4, -1)$
 (C) $(6, -3)$ (D) $(3, -3)$
14. If equations $x^3 + 2x^2 + 5x + m = 0$ and $x^3 + 2x^2 + 3x + n = 0$ has exactly one common root and product of their uncommon roots is -1 , then value of $m + n$ is -
- (A) 0 (B) 1
 (C) -1 (D) 2
15. The number of natural solutions of the equation, $27x^9 + \left(\frac{27}{195}\right)x^6 + \left(\frac{9}{195}\right)x^3 + \frac{3}{195} = 1$, is
- (A) 0 (B) 1
 (C) 3 (D) 5
16. The number of different values of x which satisfy the equation $x + \sqrt{x^2 + \sqrt{x^3 + 1}} = 1$ is
- (A) 1 (B) 0
 (C) 2 (D) 3

SEC-B : (Maximum Marks: 40)

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

17. Two AMs A_1 and A_2 , two GMs G_1 and G_2 and two HM's H_1 and H_2 are inserted between two given non-zero real numbers x and y . Then

$$\frac{1}{H_1} + \frac{1}{H_2} =$$

(A) $\frac{1}{A_1} + \frac{1}{A_2}$

(B) $\frac{1}{G_1} + \frac{1}{G_2}$

(C) $\frac{G_1 G_2}{A_1 + A_2}$

(D) $\frac{A_1 + A_2}{G_1 G_2}$

18. If $S_p = \frac{1^2}{1.2} + \frac{1^2 + 2^2}{2.3} + \frac{1^2 + 2^2 + 3^2}{3.4} + \dots + p$ terms and if $S_p + S_q - S_r + \frac{1}{6} = 0$, then smallest possible value of $p + q + r$ is -

(A) 9

(B) 12

(C) 16

(D) 25

19. The sum to 50 terms of the series $1 + 2\left(1 + \frac{1}{50}\right) + 3\left(1 + \frac{1}{50}\right)^2 + \dots$ is given by

(A) 2500

(B) 2550

(C) 2450

(D) None of these

20. Value of

$$\frac{\sin 20^\circ + \sin 140^\circ + \sin 200^\circ}{(\sin 10^\circ + \sin 70^\circ + \sin 100^\circ)(\sin 5^\circ \sin 35^\circ \sin 50^\circ)}$$

(A) 2

(B) 4

(C) 8

(D) 16

1. Let ABC and ABC' be two non-congruent triangles with sides $AB = 8$, $AC = AC' = 4\sqrt{2}$ and angle $B = 30^\circ$. The absolute value of the difference between the areas of these triangles is λ then $\frac{\lambda}{2}$ is :

2. If $\operatorname{cosec}^6 \theta + \sin^6 \theta = 3m - m^3$; $\theta \neq n\pi$ and $\frac{m^3 - m^2 - 2m}{m^2 + m - 56} \geq 0$, then number of integral values of 'm' is

3. If α, β ($\alpha < \beta$) are the two roots of the equation $\frac{1 - 8(\log_{10} x)^2}{\log_{10} x - 2(\log_{10} x)^2} = 1$ and $K = \frac{\beta^3 + 1}{\alpha^2 \beta^3}$, then sum of digits of K is

4. If locus of a point P, inside the triangle formed by the lines $y - 2x = 0$, $2x + 4y = 1$ and x-axis, which moves such that sum of its perpendicular distances from three sides of triangle is unity, is $2x + ay = b$, then $b - a$ is

5. If $ax^2 - 2hxy - ay^2 = 0$ represents a pair of straight lines forming with $2x + 3y = 6$ an isosceles right angled triangle at the origin then least value of $h - a$ is equal to (where $h, a \in \mathbb{I}$)
6. Consider the family F of the straight lines passing through the intersection of the lines $2x + y + 3 = 0$ and $5x + 4y - 9 = 0$. Then number of lines belonging to the family F whose distance from the point $(13, -10)$ is in the form $\left(28 + \alpha + \frac{1}{\alpha}\right)$, where α is a parameter ($\alpha \in \mathbb{R}^+$) is
7. The point (x, y) lies on line $3x + 4y = 6$, such that the smallest value of $25|x^2 - 2x + y^2 + 1|$ is m , then the value of $\frac{m}{3}$ is
8. Through a given point $P(5, 2)$, secants are drawn to cut the circle $x^2 + y^2 = 25$ at points $A_1(B_1), A_2(B_2), A_3(B_3), A_4(B_4)$ and $A_5(B_5)$ such that $PA_1 + PB_1 = 5, PA_2 + PB_2 = 6, PA_3 + PB_3 = 7, PA_4 + PB_4 = 8$ and $PA_5 + PB_5 = 9$. Find the value of $\sum_{i=1}^5 PA_i^2 + \sum_{i=1}^5 PB_i^2$.
[Note : $Ar(Br)$ denotes that the line passing through $P(5, 2)$ meets the circle $x^2 + y^2 = 25$ at two points A_r and B_r .]
9. Let P and Q are two distinct points on circle $x^2 + y^2 + 2gx + 2fy + c = 0$ such that circle has minimum radius. Consider P and Q are two points from which we can draw perpendicular tangents on both circles $x^2 + y^2 - 32 = 0$ and $x^2 + y^2 - 4x - 4y - 10 = 0$ then the value of $g + f + c$ is
10. If $x = \sqrt{\tan \theta} - 1$ and $y = \sqrt{\frac{\cos \theta - \sin \theta}{\sin \theta + \cos \theta}} + 2$, then the value of $(2 + 2x + x^2)(5 - 4y + y^2)$ is -

Space for Rough Work